Lower body routine — knee rehab phase

Knee-sensitive · hypermobility-safe · clean antagonist pairing · 21 working sets · no single-leg eccentric loading

Muscle group

Quads · Hamstrings · Glutes · Calves

30–60 min before: collagen peptides (10–15g) + vitamin C
Supports tendon and cartilage loading during session
Warmup activation handled by mobility protocol (banded TKEs · windshield wiper iso · lateral walks)

A — Superset 1: Hinge + Quad iso (~8 min · 3 rounds · 90s rest after each pair)

1 **Dumbbell Romanian deadlift (bilateral · soft knee bend throughout)** — 3 × 10–12
Hamstrings · glutes · erector spinae — DB · 3s ecc
Hip hinge pattern · DBs slide along thighs · stop at mid-shin · 3s eccentric lowering

↕ then immediately

2 **Wall sit (isometric)** — 3 × 20s
Quadriceps · VMO (glutes/hamstrings as isometric stabilizers only) — Bodyweight
Start shallow · only deepen if pain-free · isometric = no concentric fatigue stacking with RDL

B — Superset 2: Hip extension + Knee extension (~8 min · 3 rounds · 90s rest after each pair)

3 **Hip thrust (bench-supported)** — 3 × 8–12
Gluteus maximus · hamstrings — Bodyweight / DB
Stop just short of full hip extension · no hyperextension · drive through heels · squeeze at top

↕ then immediately

4 **Sit-to-stand — box squat variation** — 3 × 10–15
Quadriceps (dominant) · glutes (synergist, modest) — Bodyweight
Bench at 90° knee · bilateral · 3s lowering · replaces step-downs · no single-leg load

C — Superset 3: Knee flexion + Ankle (~6 min · 3 rounds · 60s rest after each pair)

5 **Prone hamstring curl** — 3 × 12
Biceps femoris · semimembranosus · semitendinosus — Band / light DB
Lying flat · light weight or resistance band · no knee snap at bottom · replaces Nordic curls

↕ then immediately

6 **Standing calf raises (hold wall or rack for balance · controlled lowering)** — 3 × 10–15
Gastrocnemius (primary) · soleus (synergist) — Bodyweight
Use hand support — hypermobility caution

D — Finisher: Soleus (~3 min · straight sets)

7 **Seated calf raise** — 3 × 12–15
Soleus — deeper calf muscle, complements standing raise (gastrocnemius) — DB / plate on knees
Seated on bench, weight on knees (DB or plate) · controlled raise and lower · soft lockout

Within 60 min after: protein 30–40g + vitamin C
Supports muscle + tendon remodeling · aim 1.6–2g protein/kg/day total

Every rep, every set — hypermobility rules

- Never lock out knees — always keep a soft bend at end range
- No single-leg eccentric loading during knee rehab phase
- Skip session if joints feel unusually loose that day — reschedule, don't push
- Stop set early if any joint feels unstable — prioritize joint feel over rep count
- No pain rule — stop immediately if any knee pain or swelling
- Avoid knee hyperextension on all exercises
- Hip abductor/adductor machine: deferred until knee is stable

Session timeline

Mobility pre-work (~14 min · separate doc) → Superset A (8 min) → B (8 min) → C (6 min) → Finisher D (3 min)
Total working session: ~25 min · 21 working sets · then proceed to post-workout mobility protocol

Working set count by muscle group

Quads: 3 concentric (sit-to-stand) + 3 isometric (wall sit) — concentric kept low for knee rehab
Glutes: 3 sets direct (hip thrust) + significant synergist work from RDL — effective volume ~5–6 sets
Hamstrings: 6 sets (DB Romanian deadlift 3 + prone hamstring curls 3)
Calves: 6 sets (standing raises 3 + seated raises 3)

v4 — warmup A removed (covered by mobility protocol) · sections A/B/C/D · exercises renumbered 1–7 · 21 working sets · clean antagonist pairing